Modeling with partial differential equations
Part I: ordinary differential equations

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Definition of solutions to ordinary differential equations

Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be a continuous function (vector field)
A solution x to the ordinary differential equation

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases}$$

is a couple:

- a time interval $J \subset \mathbb{R}$ (s.t. $0 \in J$),
- a $\mathcal{C}^1$ function $x : J \to \mathbb{R}^d$,

such that the equation $\frac{dx(t)}{dt} = f(x(t))$ is satisfied for all $t \in J$

If $J = \mathbb{R}$, x is a global solution

Outline

- Modeling with ODEs
- Existence and uniqueness
- Flow, orbit, phase portrait, and stationary points
- Numerical approximations

Modeling with Ordinary Differential equations

Three examples:

- Newton's equations of motion
- Chemical kinetics
- Population dynamics

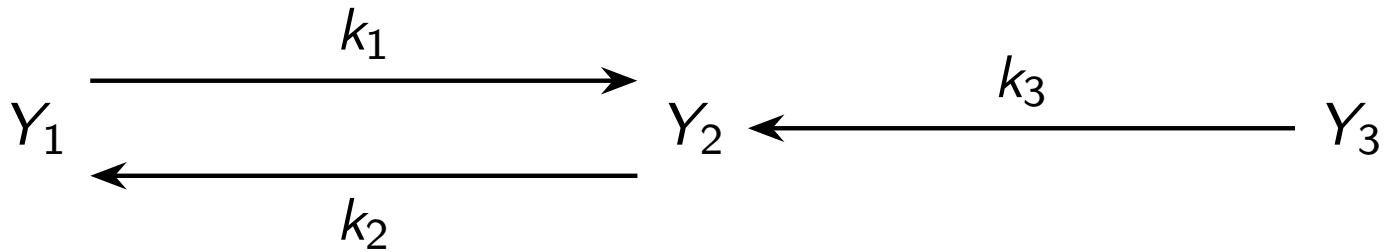
Examples

- Newton's equations of motion $m\ddot{x} = F(x)$ which can be written as:

$$\begin{cases} \dot{x} = v \\ \dot{v} = \frac{1}{m}F(x) \end{cases}$$

  Example: if $F(x) = -\nabla V(x)$, then $\frac{m|v|^2}{2} + V(x)$ is conserved along a trajectory.

- Chemical kinetics. Example:



  Concentrations $(y_i)_{1\leq i\leq 3}$ of the three species $(Y_i)_{1\leq i\leq 3}$ satisfy:

$$\begin{cases} \dot{y_1} = -k_1 y_1 + k_2 y_2 \\ \dot{y_2} = k_1 y_1 - k_2 y_2 + k_3 y_3 \\ \dot{y_3} = -k_3 y_3 \end{cases}$$

Examples

- Population dynamics. Example: Lotka-Volterra equations

$$\begin{cases} \dot{N} = N(a - bP) \\ \dot{P} = P(cN - d) \end{cases}$$

Density of prey: N. Density of predator: P
Four positive coefficients: a, b, c, d
$\rightarrow$ Some simulations...

Questions we would like to address:

- Does there exist a unique solution?
- What can be said on the behavior of this solution (periodic, convergence to a stationary point)?
- How to simulate the dynamics on a computer?

Existence and uniqueness

- Catalogue of analytically solvable ODEs
- Existence and uniqueness when f is globally Lipschitz
- Existence and uniqueness when f is locally Lipschitz

Catalogue of analytically solvable ODEs (1/3)

Scalar linear ODEs ($x(t) \in \mathbb{R}$):

$$\begin{cases} \dot{x}(t) = ax(t) \\ x(0) = x_0 \end{cases} \rightarrow \forall t \in \mathbb{R},\ x(t) = x_0 \exp(at)$$

$$\begin{cases} \dot{x}(t) = ax(t) + b \qquad a \neq 0 \\ x(0) = x_0 \end{cases}$$

$$\rightarrow \forall t \in \mathbb{R},\ x(t) = x_0 \exp(at) + \frac{b}{a}(\exp(at) - 1)$$

$$\begin{cases} a\ddot{x}(t) + b\dot{x}(t) + cx(t) = 0 \qquad a \neq 0 \\ (x(0), \dot{x}(0)) = (x_0, \dot{x}_0) \end{cases}$$

Consider the roots $\lambda_\pm \in \mathbb{C}$ of $a\lambda^2 + b\lambda + c = 0$

If $\lambda_+ \neq \lambda_-$: $\rightarrow \forall t \in \mathbb{R},\ x(t) = \alpha \exp(\lambda_+ t) + \beta \exp(\lambda_- t)$

If $\lambda_+ = \lambda_-$: $\rightarrow \forall t \in \mathbb{R},\ x(t) = (\alpha + \beta t) \exp(\lambda_+ t)$

(α, β) **are determined by** $(x(0), \dot{x}(0)) = (x_0, \dot{x}_0)$

Catalogue of analytically solvable ODEs (2/3)

Vectorial linear ODEs ($x(t) \in \mathbb{R}^d$ and $A \in \mathbb{R}^{d \times d}$)

$$\begin{cases} \dot{x}(t) = Ax(t) \\ x(0) = x_0 \end{cases}$$

$\rightarrow \forall t \in \mathbb{R}$, $x(t) = \exp(tA)x_0$ where

$$\exp(tA) = \sum_{n \geq 0} \frac{(tA)^n}{n!}$$

Duhamel formula

$$\begin{cases} \dot{x}(t) = Ax(t) + b(t) \\ x(0) = x_0 \end{cases}$$

$\rightarrow \forall t \in \mathbb{R}$, $x(t) = \exp(tA)x_0 + \int_0^t \exp((t-s)A)b(s)\,ds$

Catalogue of analytically solvable ODEs (3/3)

The solution may not be global:

$$\begin{cases} \dot{x}(t) = x^2(t) \text{ for } t \geq 0 \\ x(0) = x_0 > 0 \end{cases} \rightarrow \forall t \in [0, 1/x_0), x(t) = \frac{1}{1/x_0 - t}$$

There may be many solutions:

$$\begin{cases} \dot{x}(t) = \sqrt{|x(t)|} \text{ for } t \geq 0 \\ x(0) = 0. \end{cases}$$

$$\rightarrow \text{ For any } t_0 \geq 0,\ x(t) = \begin{cases} 0 \text{ for } t \in [0, t_0] \\ \dfrac{(t - t_0)^2}{4} \text{ for } t > t_0 \end{cases} \text{ is a solution}$$

Cauchy-Lipschitz theorem: the globally Lipschitz case

Theorem (Cauchy)

Let us assume that

$$f : \mathbb{R}^d \to \mathbb{R}^d$$

is globally Lipschitz, i.e.

$$\exists L > 0, \forall x, y \in \mathbb{R}^d, \|f(x) - f(y)\| \leq L\|x - y\|.$$

Then, the problem

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases}$$

admits a unique global solution (i.e. defined for all time $t \in \mathbb{R}$).

$\to$ the proof of uniqueness

Stability when f is globally Lipschitz, Grönwall's inequality

Proposition

Let $f : \mathbb{R}^d \to \mathbb{R}^d$ be L-Lipschitz. Let $x : [0, T] \to \mathbb{R}^d$ be $\mathcal{C}^1$ s.t.

$$\begin{cases} \dot{x} = f(x) \text{ on } [0, T] \\ x(0) = x_0 \end{cases}$$

Let $y : [0, T] \to \mathbb{R}^d$ be $\mathcal{C}^1$ and satisfy the perturbed problem

$$\begin{cases} \dot{y} = f(y) + \varepsilon \text{ on } [0, T] \\ y(0) = y_0 \end{cases}$$

for $\varepsilon : [0, T] \to \mathbb{R}^d$ continuous. Then, for all $t \in [0, T]$,

$$\|x(t) - y(t)\| \leq e^{LT} \left(\|x_0 - y_0\| + \int_0^T \|\varepsilon(s)\| \, ds \right)$$

$\to$ proof in the exercise session

Cauchy-Lipschitz theorem: the locally Lipschitz case

Theorem (Cauchy-Lipschitz)

Let $O \subset \mathbb{R}^d$ be an open set, and assume $f : O \to \mathbb{R}^d$ is continuous and locally Lipschitz: for every $y \in O$, there exists $\delta > 0$ such that f is Lipschitz on $B(y, \delta) \cap O$.

Then for any $y \in O$, there exists $\alpha, \delta > 0$, such that for any initial condition $x_0 \in B(y, \delta)$, the ODE

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases} \tag{1}$$

admits a unique $\mathcal{C}^1$ solution $x : (-\alpha, +\alpha) \to O$.

The time span 2α is independent of $x_0 \in B(y, \delta)$

Uniqueness $\to$ we can define maximal solutions (consider the union of the intervals of definition of the solutions)

Cauchy-Lipschitz theorem: the locally Lipschitz case

A maximal solution $x : J \to O$ is a solution that cannot be extended beyond the interval J

Example: the maximal solution to $\begin{cases} \dot{x}(t) = x^2(t) \\ x(0) = x_0 \end{cases}$ is $x(t) = \frac{1}{1/x_0 - t}$ defined on $J = (-\infty, 1/x_0)$ if $x_0 > 0$ and $J = (1/x_0, +\infty)$ if $x_0 < 0$

Theorem

Let $x : J \to O$ be a maximal solution of the ODE on $J \subset \mathbb{R}$.
If $\sup J < \infty$, then $x(t)$ leaves every compact subset of O as $t \to \sup J$: $\forall K$ compact set included in O, $\exists t^* \in J$ such that $\forall t \in (t^*, \sup J)$, $x(t) \notin K$.
If $\inf J > -\infty$, then $x(t)$ leaves every compact subset of O as $t \to \inf J$

If the maximal solution is not global, it therefore "blows up on the boundary of J"

Back to the Lotka-Volterra equations

Lotka-Volterra equations

$$\begin{cases} \dot{N} = N(a - bP) \\ \dot{P} = P(cN - d) \end{cases}$$

One can check that $H(N, P) = cN - d \ln |N| + bP - a \ln |P|$ is constant along trajectories. The maximal solutions $(N(t), P(t))_{t \in J}$ live on the level set $\{(N, P),\ H(N, P) = H(N(0), P(0))\}$ of H Since $N(0) > 0$ and $P(0) > 0$, N and P live in the positive quadrant

The level sets of H are closed and bounded and thus compact. Therefore, the maximal solutions are global

On $\{(N, P),\ H(N, P) = H(N(0), P(0))\}$, the velocity $\|N(a - bP), P(cN - d)\|$ is bounded from below
$\rightarrow$ Thus, N and P are periodic in time

Flow, orbit, phase portrait, and stationary points

- Flow, orbit, phase portrait
- Stability of stationary points

Flow, Orbit

Let us consider the Cauchy problem:

$$\begin{cases} \dot{x} = f(x), \\ x(0) = x_0, \end{cases}$$

with $f : O \to \mathbb{R}^d$ of class $\mathcal{C}^1$, O open set of $\mathbb{R}^d$

The flow associated with the ODE is the map

$$\phi(x_0, t) \mapsto x(t)$$

defined for $x_0 \in O$ and t in the interval $J(x_0)$ of the maximal solution starting from x_0

One has: $\forall s, t$ (such that $s, s+t \in J(x_0)$), $\phi(\phi(x_0, s), t) = \phi(x_0, s+t)$

The orbit of x_0 is

$$\mathcal{O}(x_0) = \{\phi(x_0, t), \forall t \in J(x_0)\}$$

Orbits cannot intersect

Stationary point, Phase portrait

The phase portrait is the set of orbits partitioning O

A stationary point is a point x^* such that $f(x^*) = 0$
Of course, if x^* is stationary $\mathcal{O}(x^*) = x^*$

In practice, one draws:

- some notable orbits, along with the direction of the vector field f along these orbits
- the stationary points

Example: Lotka-Volterra

Stability of stationary points

In practice, stationary points correspond to operating points and it is crucial to analyze the stability of these operating points

A stationary point x^* is said to be asymptotically stable if:

- Stability: $\forall \varepsilon > 0, \exists \delta > 0, \forall x \in B(x^*, \delta),\ \forall t \geq 0,$

$$\|\phi(x,t) - x^*\| \leq \varepsilon$$

- Asymptotic convergence: $\exists \delta > 0, \forall x \in B(x^*, \delta),$

$$\lim_{t \to \infty} \phi(x,t) = x^*$$

Stability: the linear case

We are interested in the stability of $0 \in \mathbb{R}^d$ for the ODE: $\dot{x} = Ax$ for $A \in \mathbb{R}^{d \times d}$

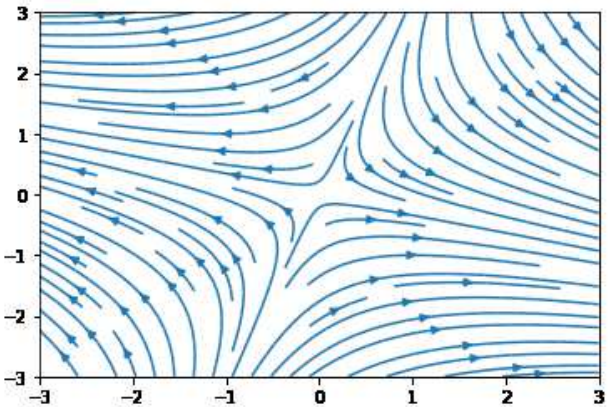
In dimension $d = 1$, $\dot{x} = ax$
$\rightarrow$ 0 is asymptotically stable iff $a < 0$.

Theorem

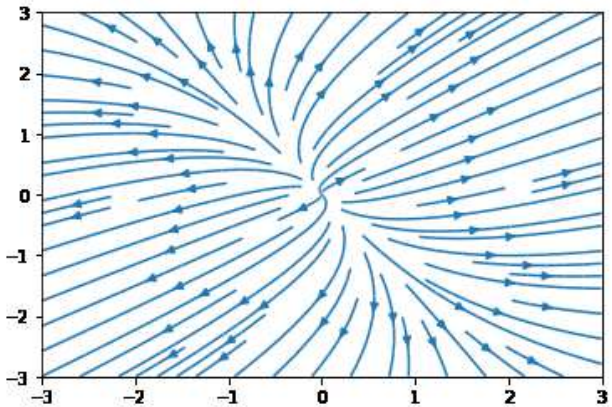
The point $x^* = 0$ is asymptotically stable if and only if all the eigenvalues of A have (strictly) negative real parts.

$\rightarrow$ proof if A is diagonalizable in $\mathbb{C}$

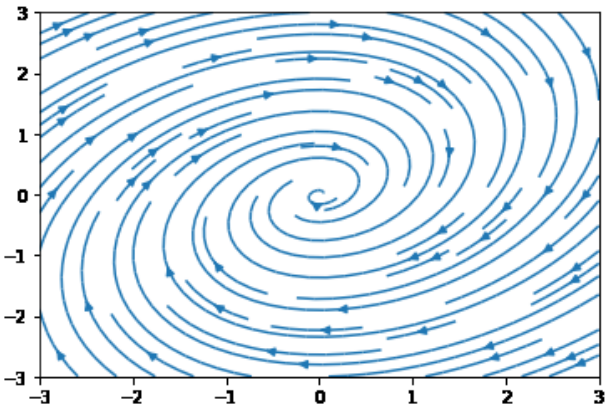
Phase portraits for $\dot{x} = Ax$, $d = 2$



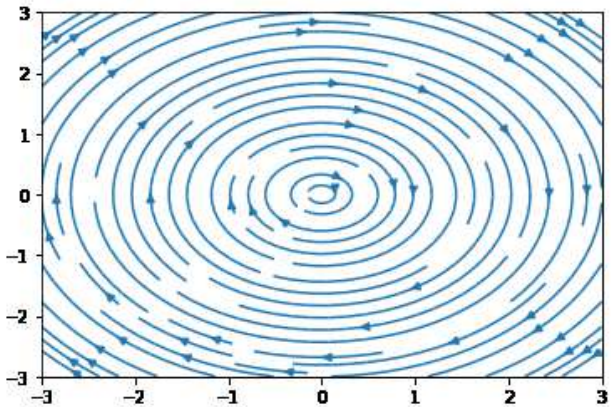
Saddle point



Repulsive node



Attractive spiral



Center

Saddle point: two real eigenvalues with opposite signs
Repulsive node: two distinct real eigenvalues with the same sign
Spiral: two complex conjugate eigenvalues with non-zero real parts
Center: two complex conjugate eigenvalues with zero real parts

Stability: the nonlinear case

Let us consider the ODE $\dot{x} = f(x)$ with a stationary point x^*, and $f : \mathbb{R}^d \to \mathbb{R}^d$ in $\mathcal{C}^1$

By Taylor expansion

$$f(x) = Df(x^*)(x - x^*) + o(\|x - x^*\|)$$

where $Df(x^*)$ is the Jacobian matrix

$$\forall i, j \in \{1, \dots, d\}, \, [Df(x^*)]_{i,j} = \frac{\partial f_i}{\partial x_j}(x^*)$$

$\to$ We are back to the linear case, locally!

Theorem

If all the eigenvalues of the matrix $Df(x^*)$ have (strictly) negative real parts, then the equilibrium point x^* is asymptotically stable
Moreover, if one of the eigenvalues has a (strictly) positive real part, then the equilibrium point x^* is unstable

Numerical approximations

- Three numerical schemes
- Convergence analysis: consistency and stability
- Absolute stability

Numerical approximations

Assume $f : \mathbb{R}^d \to \mathbb{R}^d$ is L-Lipschitz

Objective: compute an approximation for $t \geq 0$ of the unique and globally defined solution x to the Cauchy problem

$$\begin{cases} \dot{x} = f(x) \\ x(0) = x_0 \end{cases}$$

Idea:

- Introduce a time grid with time step Δt: $(t_n)_{n \geq 0}$ with $t_n = n\Delta t$
- Approximate the continuous problem using finite differences

Examples of numerical schemes

- The approximation $\frac{x(t_{n+1})-x(t_n)}{\Delta t} \approx \dot{x}(t_n) = f(x(t_n))$ gives the explicit Euler scheme:

$$x_{n+1} = x_n + \Delta t\, f(x_n)$$

- The approximation $\frac{x(t_{n+1})-x(t_n)}{\Delta t} \approx \dot{x}(t_{n+1}) = f(x(t_{n+1}))$ gives the implicit Euler scheme:

$$x_{n+1} = x_n + \Delta t\, f(x_{n+1})$$

- The approximation $\frac{x(t_{n+1})-x(t_n)}{\Delta t} \approx \dot{x}(t_{n+1/2}) = f(x(t_{n+1/2})) \approx \frac{f(x(t_n))+f(x(t_{n+1}))}{2}$ gives the Crank-Nicolson scheme:

$$x_{n+1} = x_n + \frac{\Delta t}{2}\left(f(x_n) + f(x_{n+1})\right)$$

The approximate solutions $(x_n)_{n\geq 0}$ are computed recursively starting from x_0

Implicit versus explicit

For explicit methods, x_{n+1} is obtained from x_n using a fixed (small) number of calls to f. Example: explicit Euler

For implicit methods, computing x_{n+1} requires solving an equation, typically by an iterative method which will require many calls to f. Example: implicit Euler and Crank-Nicolson

For implicit Euler, one can look for a solution to the problem

$$x_{n+1} = x_n + \Delta t\, f(x_{n+1})$$

using an iterative method ($x_{n+1} =$ " $\lim_{k\to\infty} x_{n+1}^{(k)}$ ") such as:

- Fixed point: $x_{n+1}^{(k+1)} = x_n + \Delta t\, f(x_{n+1}^{(k)})$
- Newton's method: (written here for $d = 1$)
  $x_{n+1}^{(k+1)} = x_{n+1}^{(k)} - \frac{x_{n+1}^{(k)} - x_n - \Delta t\, f(x_{n+1}^{(k)})}{1 - \Delta t\, f'(x_{n+1}^{(k)})}$

Numerical analysis

The numerical schemes we are studying are one-step methods:

$$x_{n+1} = x_n + \Delta t\, \Psi_{\Delta t}(x_n)$$

where $\Psi_{\Delta t}$ is the increment function
Example: for Euler explicit, $\Psi_{\Delta t}(x) = f(x)$

The function $\Phi_{\Delta t}(x) = x + \Delta t\, \Psi_{\Delta t}(x)$ is the numerical flow over a time interval of size Δt

Two types of questions:

- Convergence. For fixed time horizon T, take $\Delta t = T/N$. What can be said about the exact solution $(x(t))_{0 \le t \le T}$ versus the approximate solution $(x_n)_{0 \le n \le N}$ as $N \to \infty$ (i.e. $\Delta t \to 0$)?
- Absolute stability. For fixed Δt, does the approximate solution x_n behave as $n \to \infty$ like the exact solution $x(t)$ as $t \to \infty$?

Convergence analysis

Recall that T is fixed, $\Delta t = \frac{T}{N}$, $t_n = n\Delta t = \frac{nT}{N}$

We would like to analyze the error

$$\sup_{0 \leq n \leq N} |x(t_n) - x_n|$$

in the limit $N \to \infty$ i.e. $\Delta t \to 0$

Two steps (Lax principle):

- Consistency: What is the error introduced by the numerical scheme over one time step?
- Stability: How do these consistency errors accumulate over the N iterations?

Consistency

The consistency error on a time interval of size Δt and for an initial condition x_0 is:

$$\varepsilon_{\Delta t}(x_0) = \phi(x_0, \Delta t) - \Phi_{\Delta t}(x_0)$$

The numerical scheme is consistent if, in the limit $\Delta t \to 0$,

$$\varepsilon_{\Delta t}(x_0) = o(\Delta t)$$

(for any continuous Lipschitz function f, and for any x_0)

The numerical scheme is consistent of order p if, in the limit $\Delta t \to 0$,

$$\varepsilon_{\Delta t}(x_0) = O(\Delta t^{p+1})$$

(for any f of class $\mathcal{C}^p$, and for any x_0)

Consistency: the explicit Euler scheme in 1d

On the one hand $\phi(x_0, \Delta t) = x(\Delta t)$ where $\dot{x}(t) = f(x(t))$ and $x(0) = x_0$. In particular, (Taylor expansion) for some $\theta \in [0, 1]$,

$$\phi(x_0, \Delta t) = x_0 + \Delta t\, f(x_0) + \frac{\Delta t^2}{2} \ddot{x}(\theta \Delta t)$$

On the other hand,

$$\Phi_{\Delta t}(x_0) = x_0 + \Delta t\, f(x_0)$$

Thus

$$\varepsilon_{\Delta t}(x_0) = \frac{\Delta t^2}{2} \ddot{x}(\theta \Delta t) = O(\Delta t^2)$$

since, if f is $\mathcal{C}^1$, $\ddot{x}(t) = f'(x(t))f(x(t))$ is uniformly bounded on $[0, \Delta t]$.

$\rightarrow$ The explicit Euler scheme is of order 1

From consistency to error

One can check that explicit and implicit Euler are consistent of order 1, and Crank-Nicolson is consistent of order 2

If one plugs the real solution in the numerical scheme, what do we get?

$$\begin{aligned} x(t_{n+1}) &= \phi(x(t_n), \Delta t) \\ &= \Phi_{\Delta t}(x(t_n)) + \varepsilon_{\Delta t}(x(t_n)) \end{aligned}$$

The numerical scheme is: $x_{n+1} = \Phi_{\Delta t}(x_n)$.

Thus, $(x(t_n))_{n\geq 0}$ is solution to a perturbed numerical scheme. How do the errors accumulate?

$\rightarrow$ A numerical scheme is stable "if the errors do not accumulate catastrophically"

Stability

Proposition

Assume that the increment function $\Psi_{\Delta t}$ is Lipschitz, with a Lipschitz constant $\overline{L}$ independent of Δt. Then the numerical scheme is stable in the following sense: if (x_n) satisfies

$$x_{n+1} = x_n + \Delta t \Psi_{\Delta t}(x_n),$$

and (y_n) satisfies

$$y_{n+1} = y_n + \Delta t \Psi_{\Delta t}(y_n) + \varepsilon_n,$$

then for all N,

$$\max_{0 \leq n \leq N} |x_n - y_n| \leq \exp(\overline{L} T) \left(|x_0 - y_0| + \sum_{k=0}^{N-1} |\varepsilon_k| \right).$$

$\rightarrow$ proof

Stability

Application of the previous proposition: if $\Psi_{\Delta t}$ is $\overline{L}$-Lipschitz

$$x_{n+1} = x_n + \Delta t \Psi_{\Delta t}(x_n)$$

$$x(t_{n+1}) = x(t_n) + \Delta t \Psi_{\Delta t}(x(t_n)) + \varepsilon_{\Delta t}(x(t_n))$$

For all N,

$$\max_{0 \leq n \leq N} |x_n - x(t_n)| \leq \exp(\overline{L} T) \left(\sum_{k=0}^{N-1} |\varepsilon_{\Delta t}(x(t_k))| \right).$$

Convergence

ODE: $(x(t))_{0\leq t\leq T}$ solution to $\dot{x} = f(x)$
Numerical scheme: $x_{n+1} = x_n + \Delta t \Psi_{\Delta t}(x_n)$, with $\Delta t = T/N$ and $x_0 = x(0)$

Theorem

If the scheme satisfies:

- *Consistency in the following sense: there exists $K > 0$ such that for all $t \in [0, T]$, for all Δt,*

$$|\varepsilon_{\Delta t}(x(t))| \leq K \Delta t^{p+1}$$

- *Stability in the following sense: $\Psi_{\Delta t}$ is $\overline{L}$-Lipschitz*

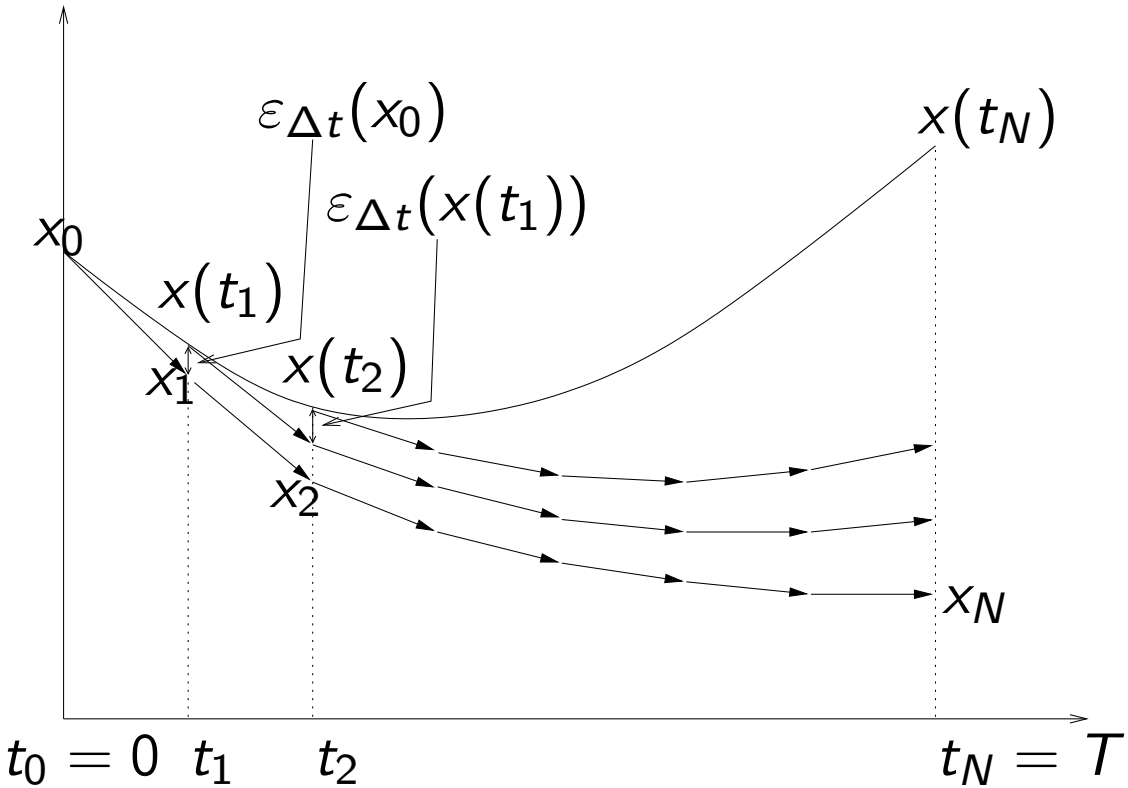
Then the scheme is convergent: there exists $C > 0$ s.t. for all Δt,

$$\sup_{0\leq n\leq N} |x(t_n) - x_n| \leq C \Delta t^p$$

$\rightarrow$ proof

Example: the explicit Euler scheme is consistent with order $p = 1$, and stable with $\overline{L} = L$

Numerical illustration of the convergence proof



A solid line indicates a solution obtained by the exact flow
An arrowed line indicates a solution obtained by applying the numerical scheme

Absolute stability

Consider the model problem:

$$\dot{x} = \lambda x$$

for $\lambda \in \mathbb{C}$, and $x(t) \in \mathbb{C}$.

One has $\lim_{t\to\infty} |x(t)| = 0$ for any $x(0)$ iff $\mathrm{Re}(\lambda) < 0$

A numerical method is absolutely stable if, when applied to $\dot{x} = \lambda x$, with λ s.t. $\mathrm{Re}(\lambda) < 0$, then, for any x_0,

$$\lim_{n\to\infty} |x_n| = 0$$

Absolute stability: examples

For explicit Euler, one has

$$x_{n+1} = x_n + \lambda \Delta t x_n = R(\lambda \Delta t) x_n = [R(\lambda \Delta t)]^n x_0$$

with $R(z) = 1 + z$.
The method is absolutely stable iff $\mathrm{Re}(\lambda) < 0$ implies $|R(z)| < 1$.
This is equivalent to

$$|1 + \lambda \Delta t| < 1$$

which imposes restrictions on Δt. The scheme is conditionally absolutely stable

Absolute stability: examples

For implicit Euler, one has $x_{n+1} = x_n + \lambda \Delta t x_{n+1}$ and thus

$$x_{n+1} = R(\lambda \Delta t) x_n = [R(\lambda \Delta t)]^n x_0$$

with $R(z) = 1/(1-z)$.
The method is absolutely stable iff $\mathrm{Re}(\lambda) < 0$ implies $|R(z)| < 1$.
This is equivalent to

$$|1 - \lambda \Delta t| > 1$$

which is true for all Δt. The scheme is unconditionally absolutely stable

Aside: back to the existence of solutions (1/2)

A convergent numerical scheme yields an existence result

Idea of the proof (which dates back to Augustin-Louis Cauchy, 1824)
If f is continuous, for $\beta > 0$, $M = \sup_{B(x_0,\beta)} \|f\| < \infty$. Let $T = \beta/M$. For $N \in \mathbb{N} \setminus \{0\}$, let $\Delta t_N = T/N$ and for $k \in \{0, \dots, N-1\}$,

$$x_{k+1}^{(N)} = x_k^{(N)} + \Delta t_N f(x_k^{(N)})$$

Let us define the linear interpolant

$$x^{(N)}(t) = x_0 + \int_0^t f(x^{(N)}(\lfloor s \rfloor_N))ds$$

where $\lfloor s \rfloor_N = \lfloor s/\Delta t_N \rfloor \Delta t_N$: $\forall k \in \{0, \dots, N\}$, $x^{(N)}(k\Delta t_N) = x_k^{(N)}$
One can check that

$$\max_{t\in[0,T]} \|x^{(N)}(t) - x_0\| \leq \beta$$

Aside: back to the existence of solutions (2/2)

The family of functions $(x^N) \in \mathcal{C}([0,T],\mathbb{R}^d)$ is:

- Uniformly bounded: $\sup_N \|x^{(N)}\|_{L^\infty([0,T])} < \infty$
- Uniformly Lipschitz: $\sup_N \left\| \frac{dx^{(N)}}{dt} \right\|_{L^\infty([0,T])} < \infty$

Therefore, by Arzelà-Ascoli theorem, there exists a subsequence N_m and a function $x \in \mathcal{C}([0,T],\mathbb{R}^d)$ such that

$$\lim_{m\to\infty} \|x^{(N_m)} - x\|_{L^\infty([0,T])} = 0$$

Conclude by passing to the limit $m \to \infty$ in the integral formulation of the ODE (dominated convergence):

$$x^{(N_m)}(t) = x_0 + \int_0^t f(x^{(N_m)}(\lfloor s \rfloor_{N_m}))\, ds$$

to get

$$x(t) = x_0 + \int_0^t f(x(s))\, ds$$